SINGAPORE UNIVERSITY OF TECHNOLOGY AND DESIGN

Term 7- Sept 2025

Nanoelectronics and Technology (01.119/99.503)-Week 8 Class 1

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04-Nov- 2025

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Outline

- E-beam lithography
- Nanoimprinting
- EUV Photolithography
- Nanohub Simulation
- Project 1- Discussion

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E-beam writing/lithography

- **Electron-beam lithography** (often abbreviated as **e-beam lithography** or **EBL**) is the practice of scanning a focused beam of electrons to draw custom shapes on a surface covered with an electron-sensitive film called a resist (exposing).

- One of the most commonly used methods to pattern structures on a nanometer scale.
- Special electron beam sensitive resists have to be used for EBL. The most common one is polymethyl methacrylate (PMMA).

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- The electron beam changes the solubility of the resist, enabling selective removal of either the exposed or non-exposed regions of the resist by immersing it in a solvent (developing). The purpose, as with photolithography, is to create very small structures in the resist that can subsequently be transferred to the substrate material, often by etching

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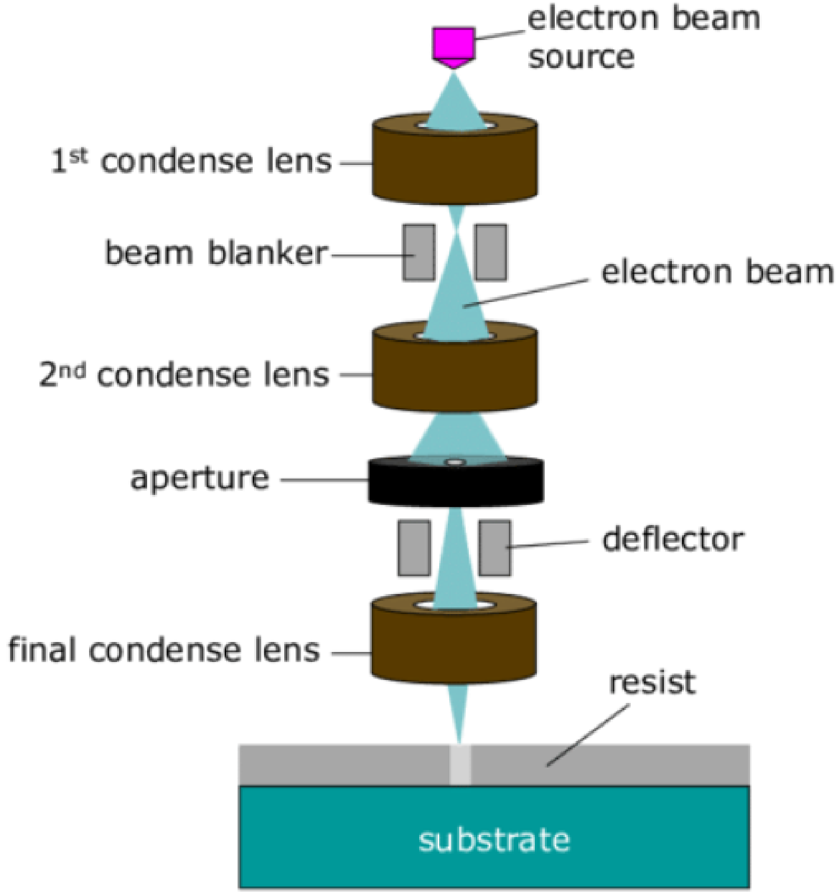
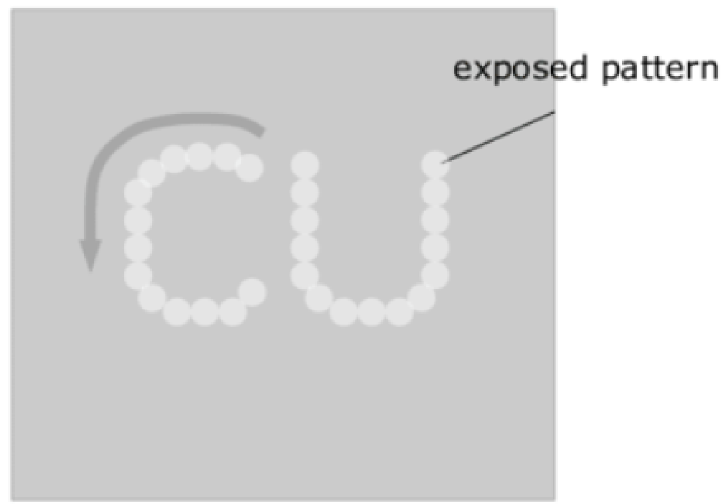
E-beam writing/lithography



Can use magnetic or electric fields for lenses



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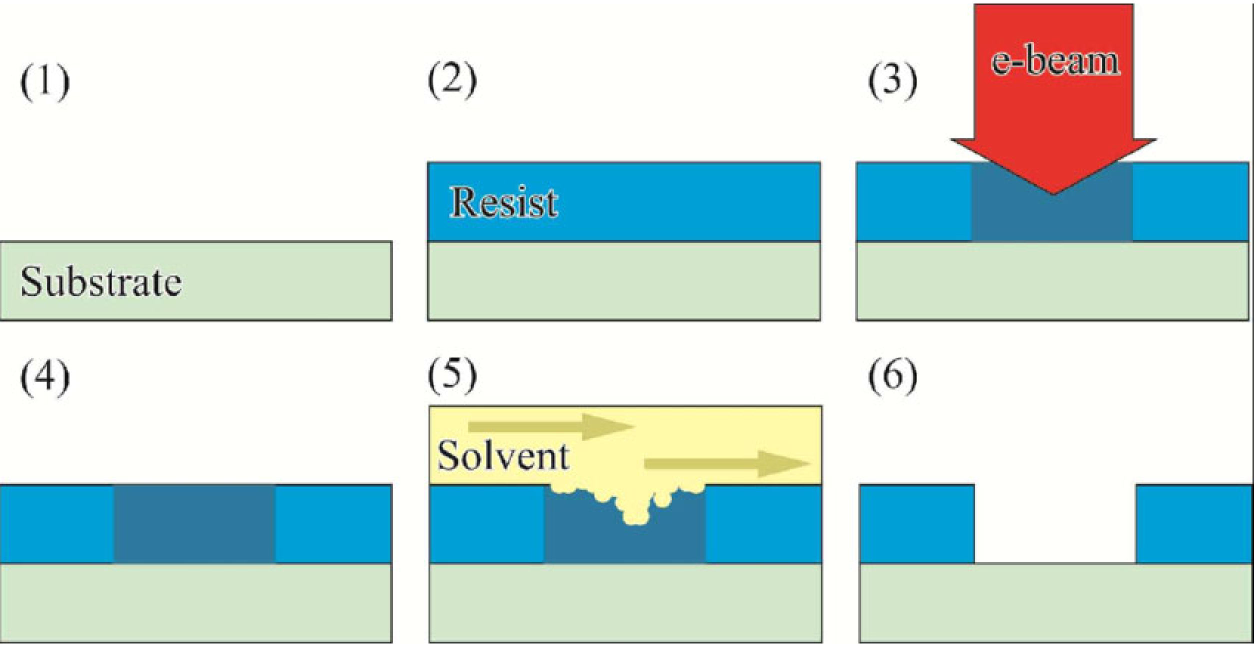
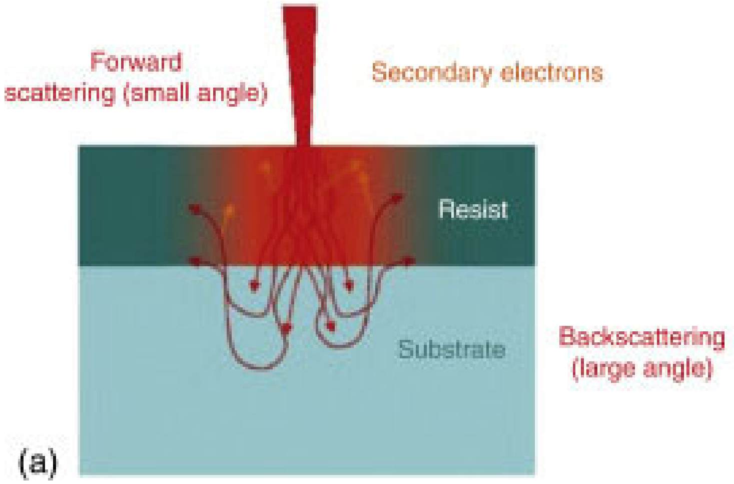
E-beam writing/lithography





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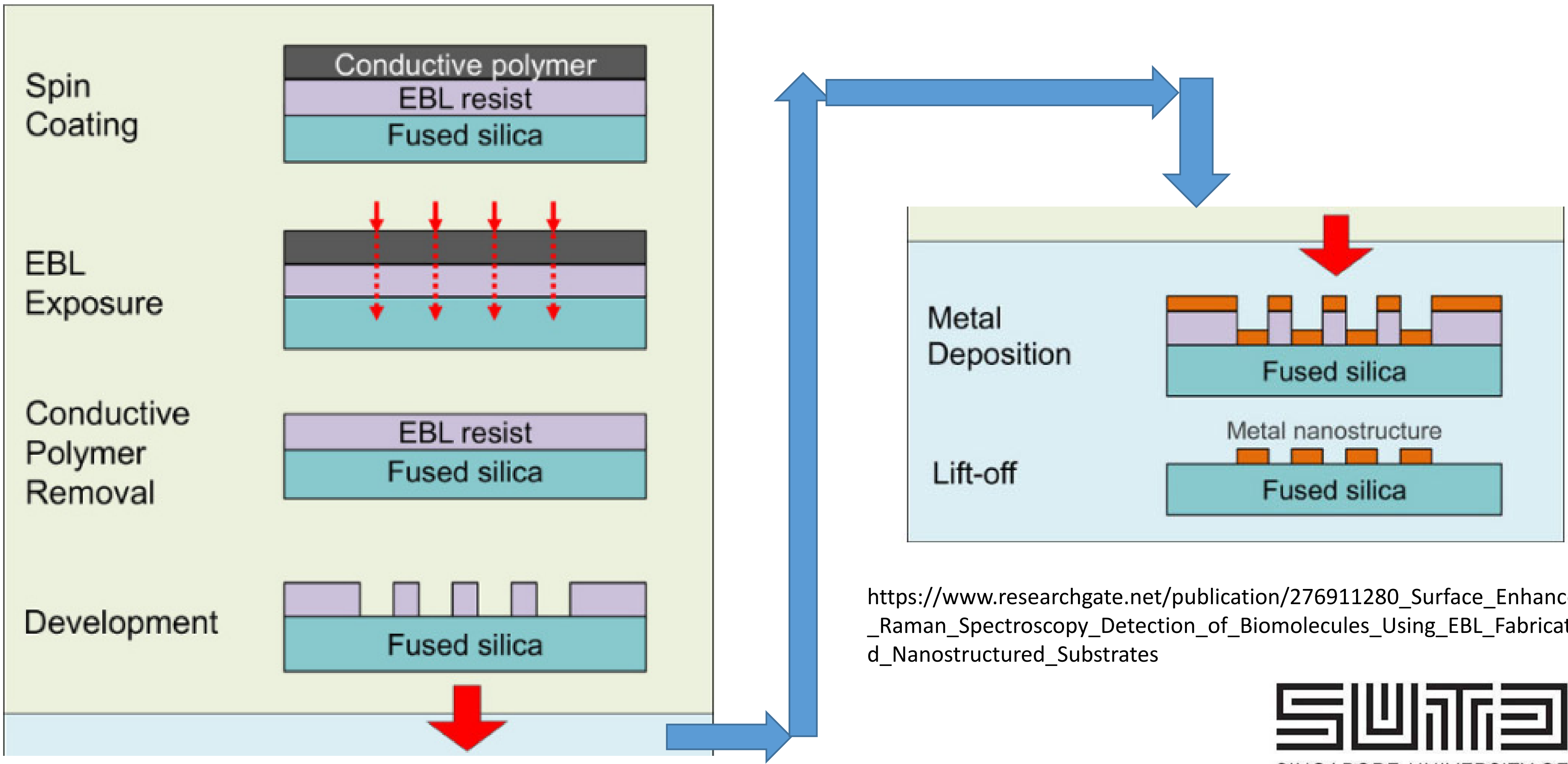
E-beam writing/lithography



https://www.researchgate.net/publication/276911280_Surface_Enhanced_Raman_Spectroscopy_Detection_of_Biomolecules_Using_EBL_Fabricated_Nanostructured_Substrates

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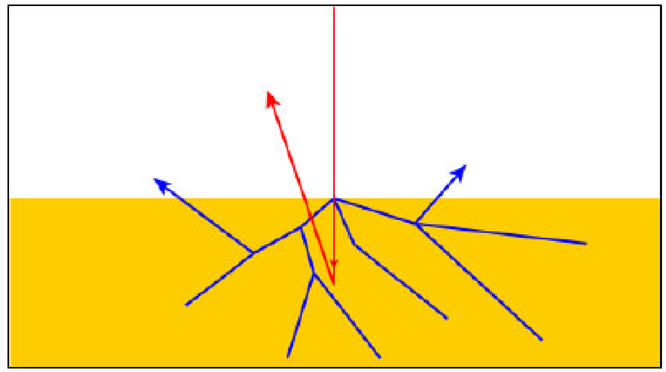
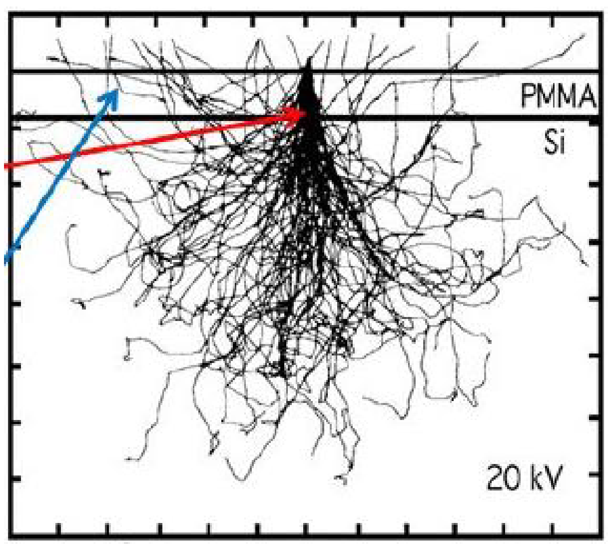
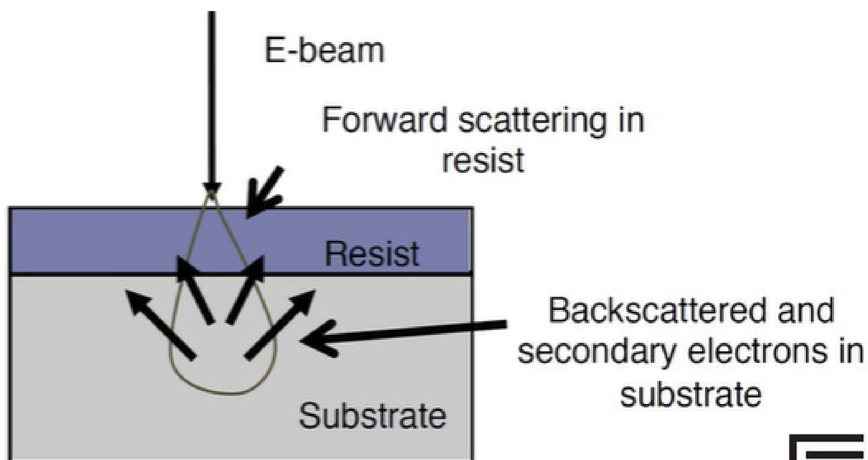
E-beam writing/lithography

Resolution limits from electron scattering:

- Forward scattering (in resist)
- Backscattering (from substrate)







REF:Electron Beam Lithography, Adam Ramm

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E-beam writing/lithography

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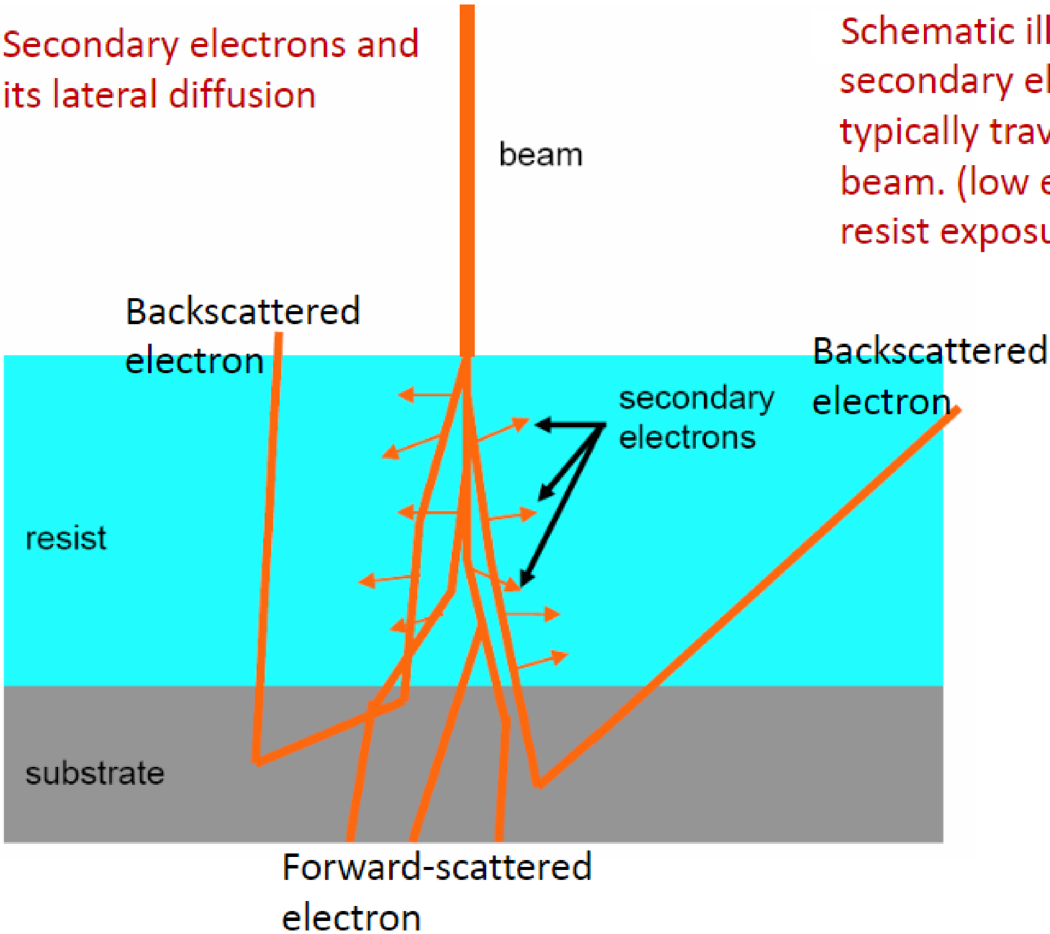
E-beam writing/lithography

Secondary electrons and its lateral diffusion



Schematic illustration of secondary electrons (SE), which typically travel/diffuse normal to the beam. (low energy SE is responsible for resist exposure)

https://studylib.net/doc/9599457/electron-beam-lithography_3

Whether it is forward scattering or beam spot size that limits the minimum line-width, the effect of SE diffusion is to further broaden the line/reduce the resolution.

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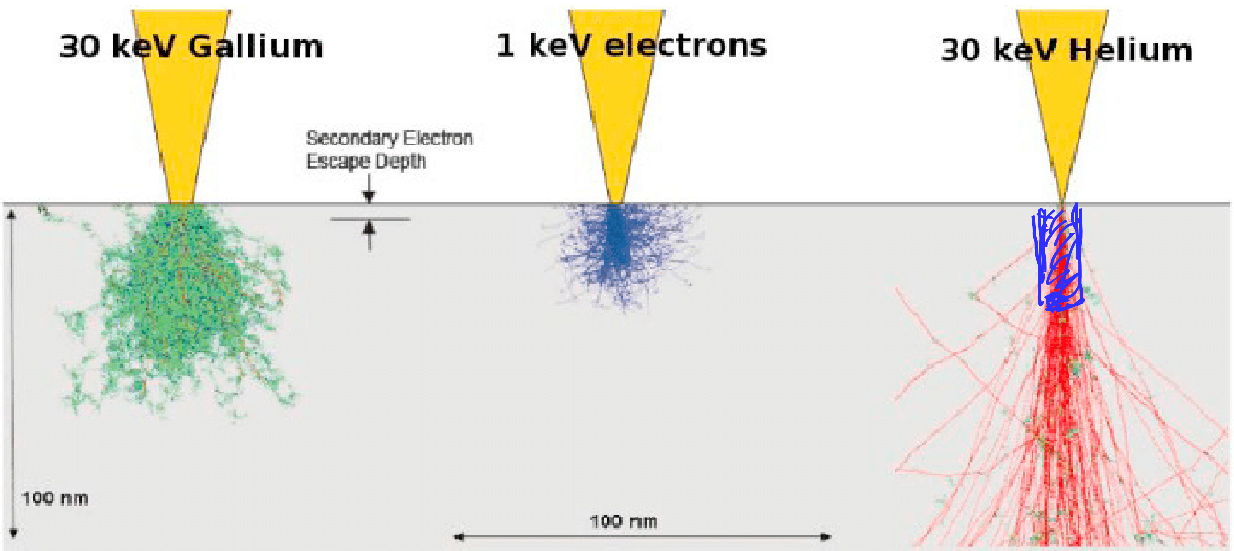
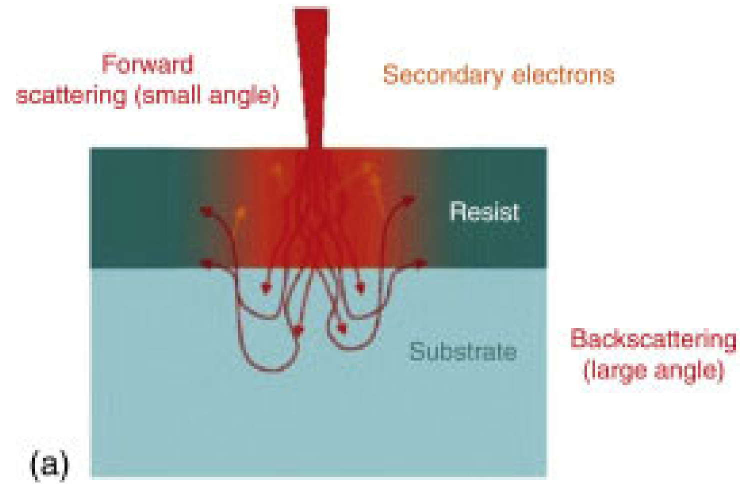
E-beam writing/lithography





https://www.sciencedirect.com/topics/engineering/electron-optical-lithography

DOI:10.1016/j.promfg.2019.02.038

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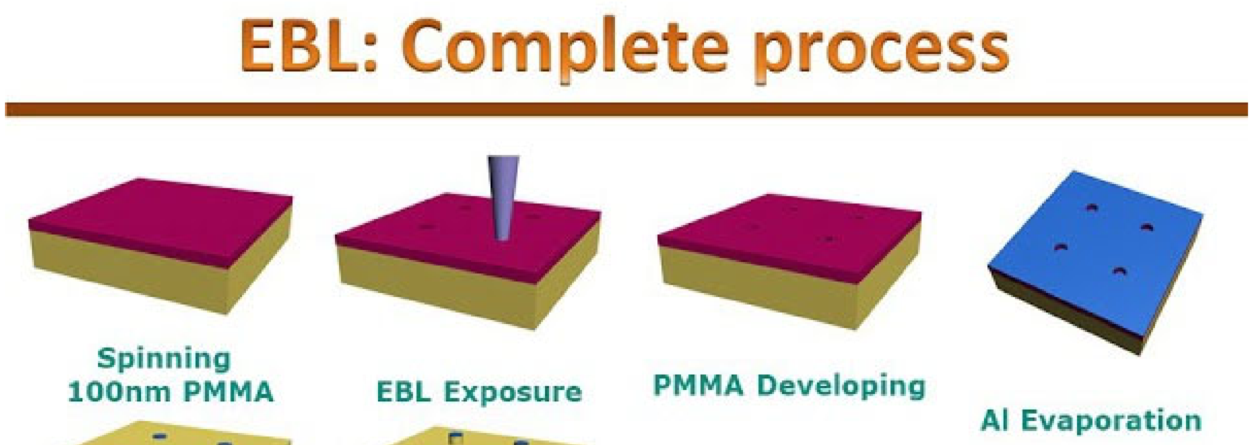
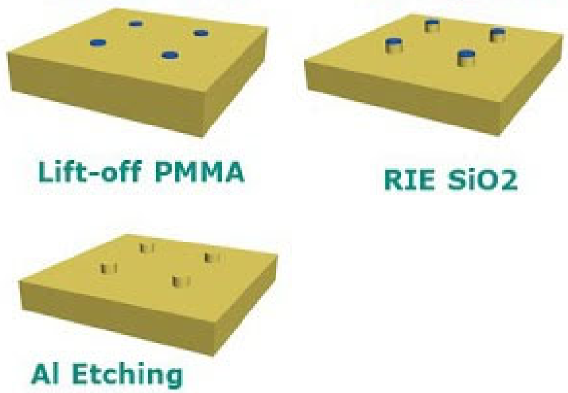
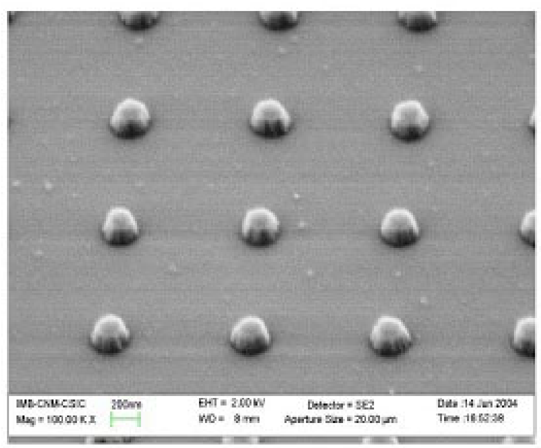
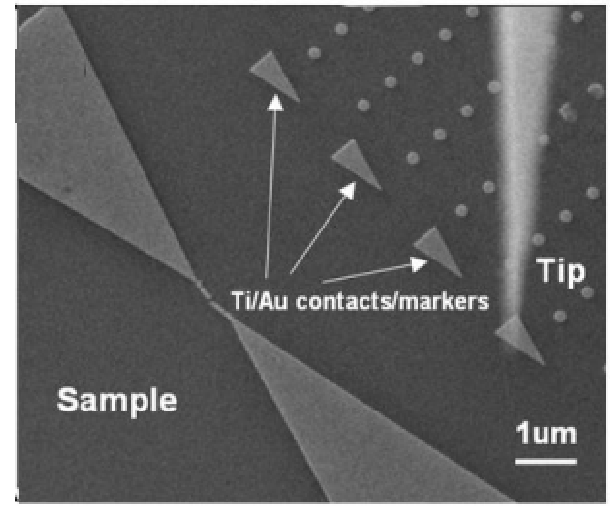
E-beam writing/lithography









F. Torres, UA De Barcelona-Electron beam lithography

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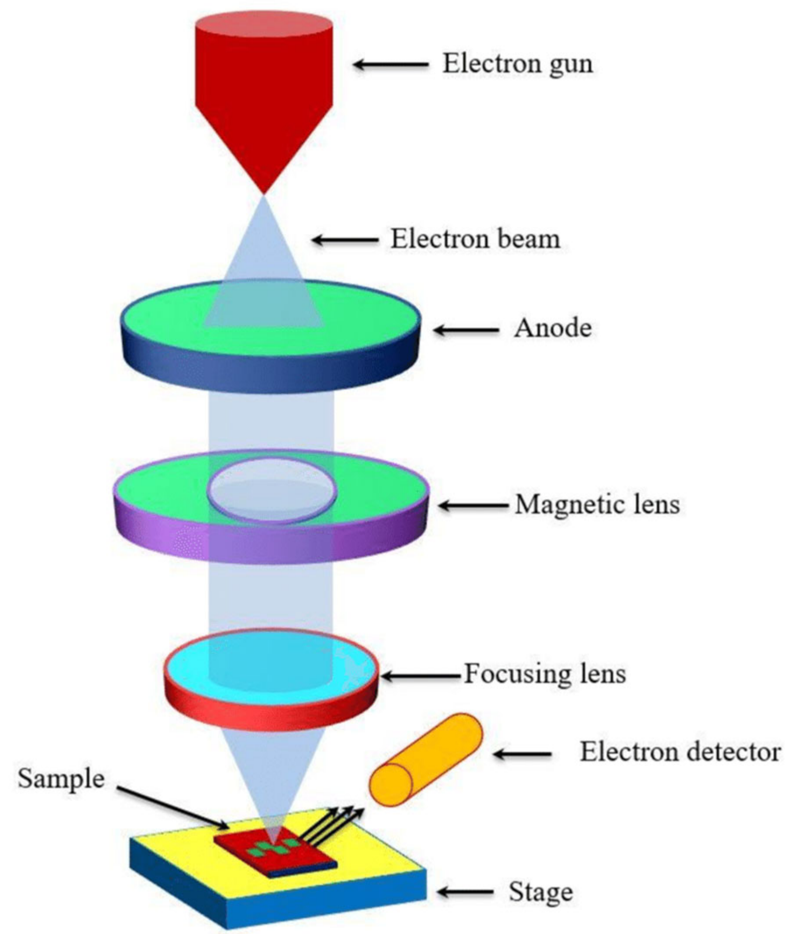
E-beam writing/lithography

- E-beam resist **Polymethyl methacrylate** (PMMA).
- Spin coating
- No mask required
- Slow process
- High resolution
- Suitable for custom based pattern
- Electron scattering is the main limitation of writing small features.



Pijush MS thesis, 2016
https://www.researchgate.net/figure/Schematic-diagram-of-the-E-beam-lithography_fig3_322317317

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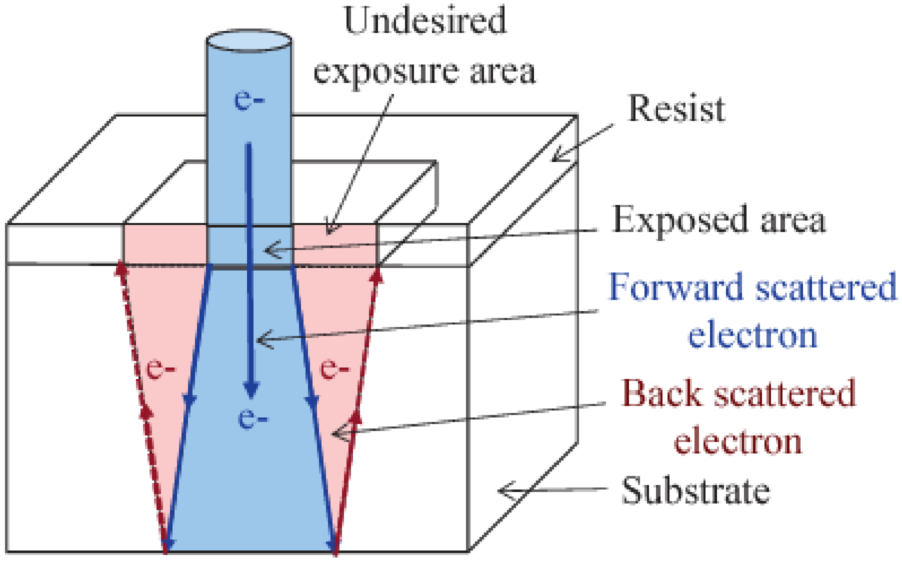
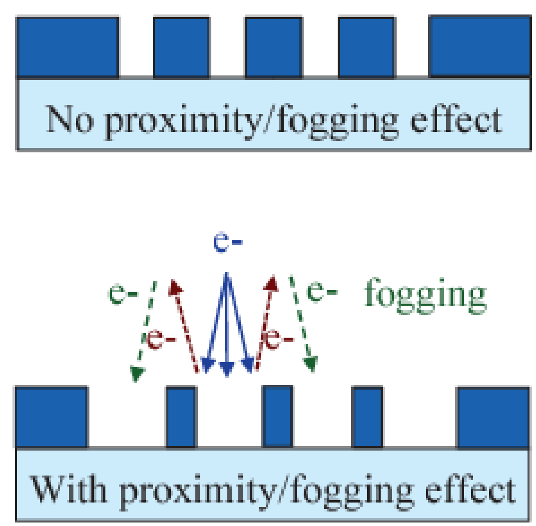
E-beam writing/lithography- Proximity Effect

The proximity effect in electron beam lithography (EBL) is the phenomenon that the exposure dose distribution, and hence the developed pattern, is wider than the scanned pattern due to the interactions of the beam electrons (scattered) with the resist and substrate.



(a)



(b)

•DOI:10.1145/2744769.2747925

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- A limiting factor of high resolution and contrast of EBL.
- Depends on the pattern density and the substrate material, as well as parameters of the EBL exposure.
  - Acceleration voltage
  - Electron dose

https://slideplayer.com/slide/8510880/

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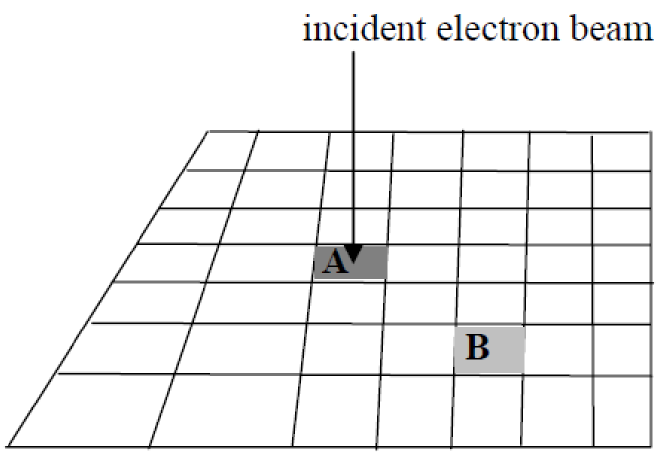
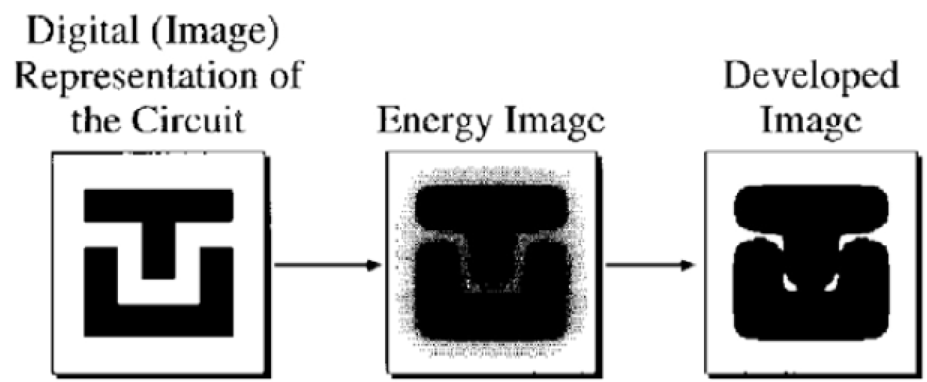


Proximity effect: exposure at pixel A affects pixel B.



Proximity Effect in E-beam Lithography By: Araldo van de Kraats

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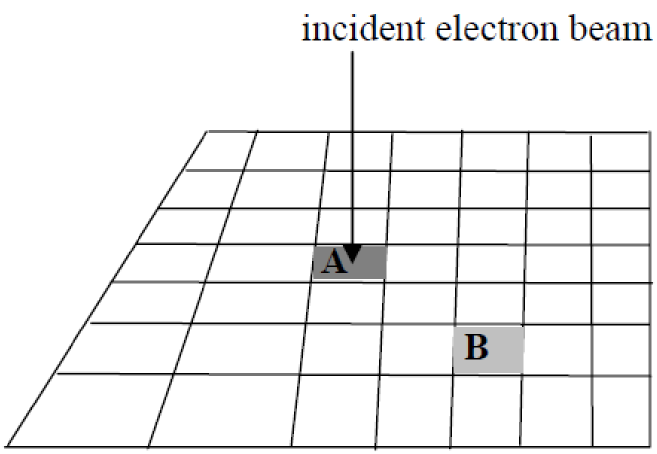

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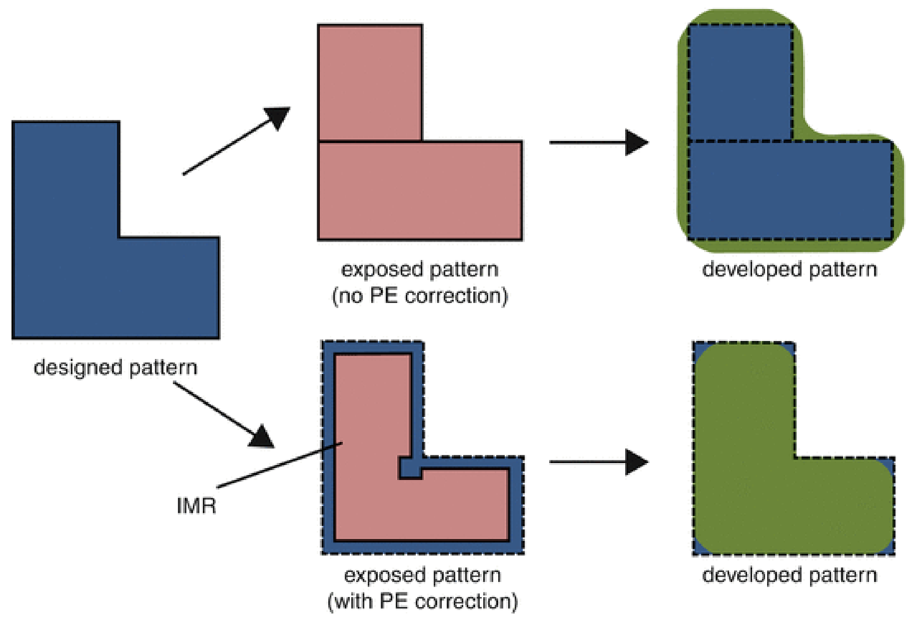


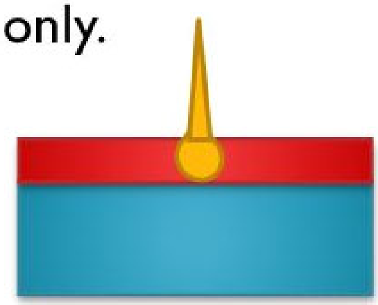
Proximity Effect in E-beam Lithography By: Araldo van de Kraats

https://link.springer.com/referenceworkentry/10.1007/978-90-481-9751-4_344

E-beam writing/lithography

Reduction of proximity effect

- **Low energy EBL**: (1-3 keV)
  - Dissipation of electron energy in the resist only.
  - No generation of secondary electrons in the substrate!



- **Limitations:**
  - Forward scattering is large -> low resolution.
  - E-beam size is big due to Column interaction between electrons.
  - Electron optics works poor at low energy.
  - Resist must be thin for complete exposure (e.g. 70 nm for 2 keV) difficult to use.
  - Hard to focus the e-beam, the beam is very sensitive to external fields

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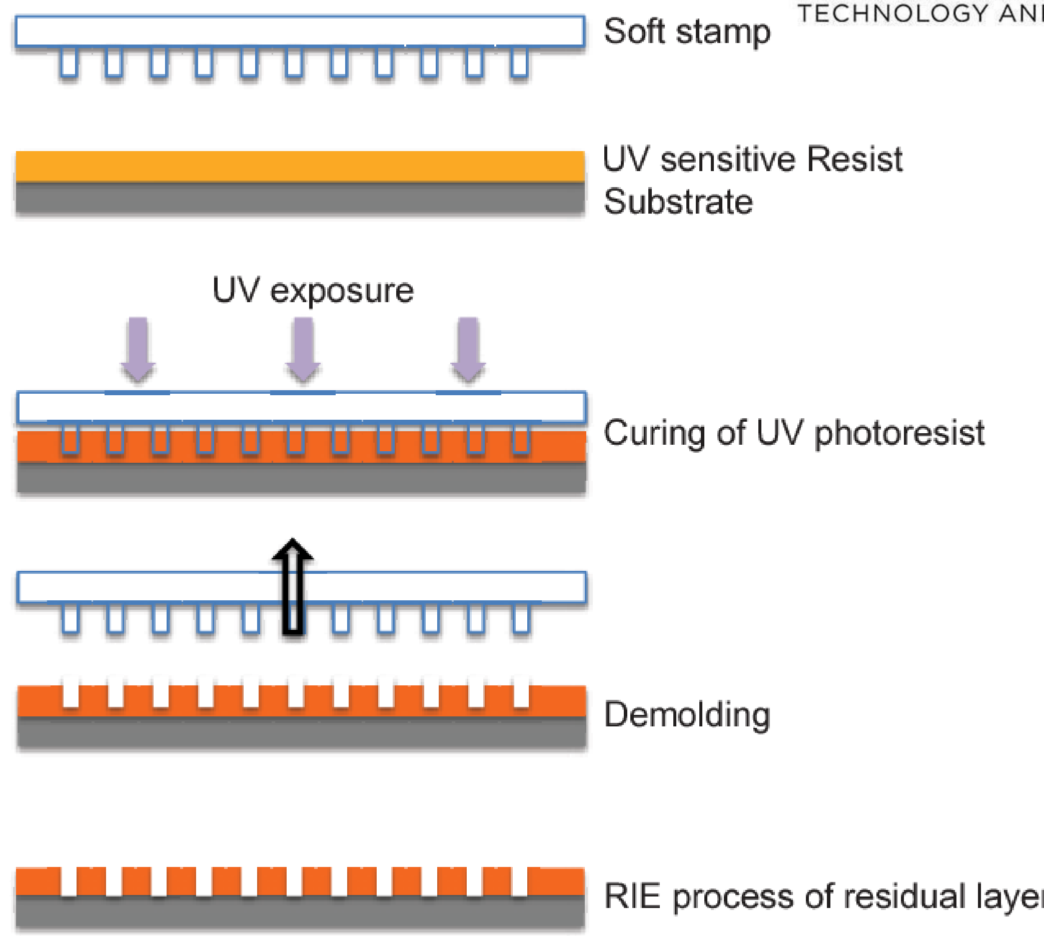
E-beam writing/lithography

- **Advantages**
  - Extremely high resolution
  - Direct patterning on a substrate with high degree of automation (No mask required)

- **But:**
  - Low throughput (Raster Scan)
  - Not suitable for mass production
  - Proximity effect

Nanoimprinting

- Nanoimprint lithography (NIL) is **a method of fabricating nanometer scale patterns**.
- It is a simple nanolithography process with low cost, high throughput and high resolution.
- It creates patterns by mechanical deformation of imprint resist and subsequent processes.



Soft UV Nanoimprint Lithography: A Tool to Design Plasmonic Nanobiosensors-G. Barbillon, 2011

Nanoimprinting

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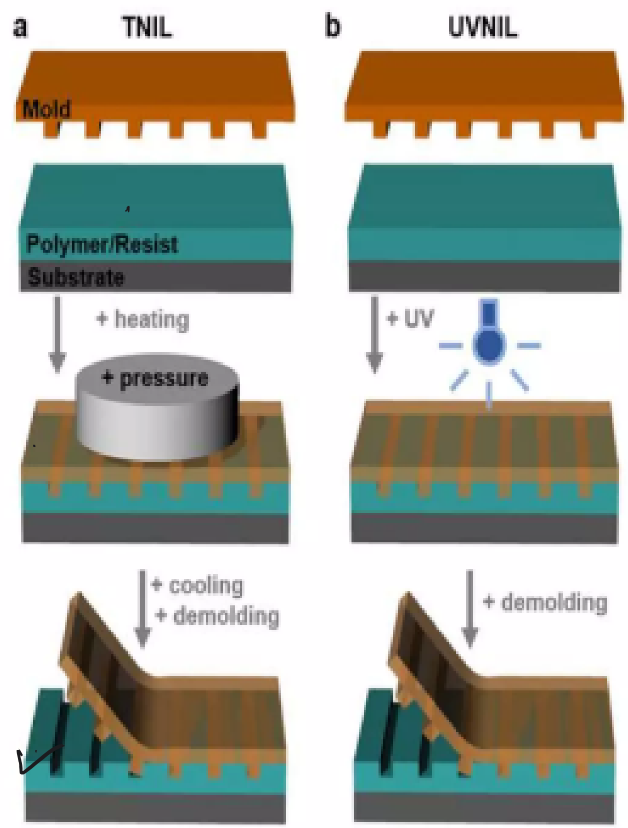
- NIL works on the principle of mechanical deformation of resists using mold containing nanostructure during heat or UV curing process.
- Mold or template is generally prepared by E-beam lithography. Materials used like Si, SiO2, Quartz, Ni
- Thermoplastic or UV curable resist are used.
- Pattern transferred by anisotropic etching to remove residue resist.



Nanoimprint Lithography

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Nanoimprinting

- NIL relies on the stamp making uniform contact with the substrate, which is challenging when dealing with surfaces that are not perfectly flat. The presence of defects in the templates or during the imprinting process can lead to pattern imperfections, affecting the quality of the final product.

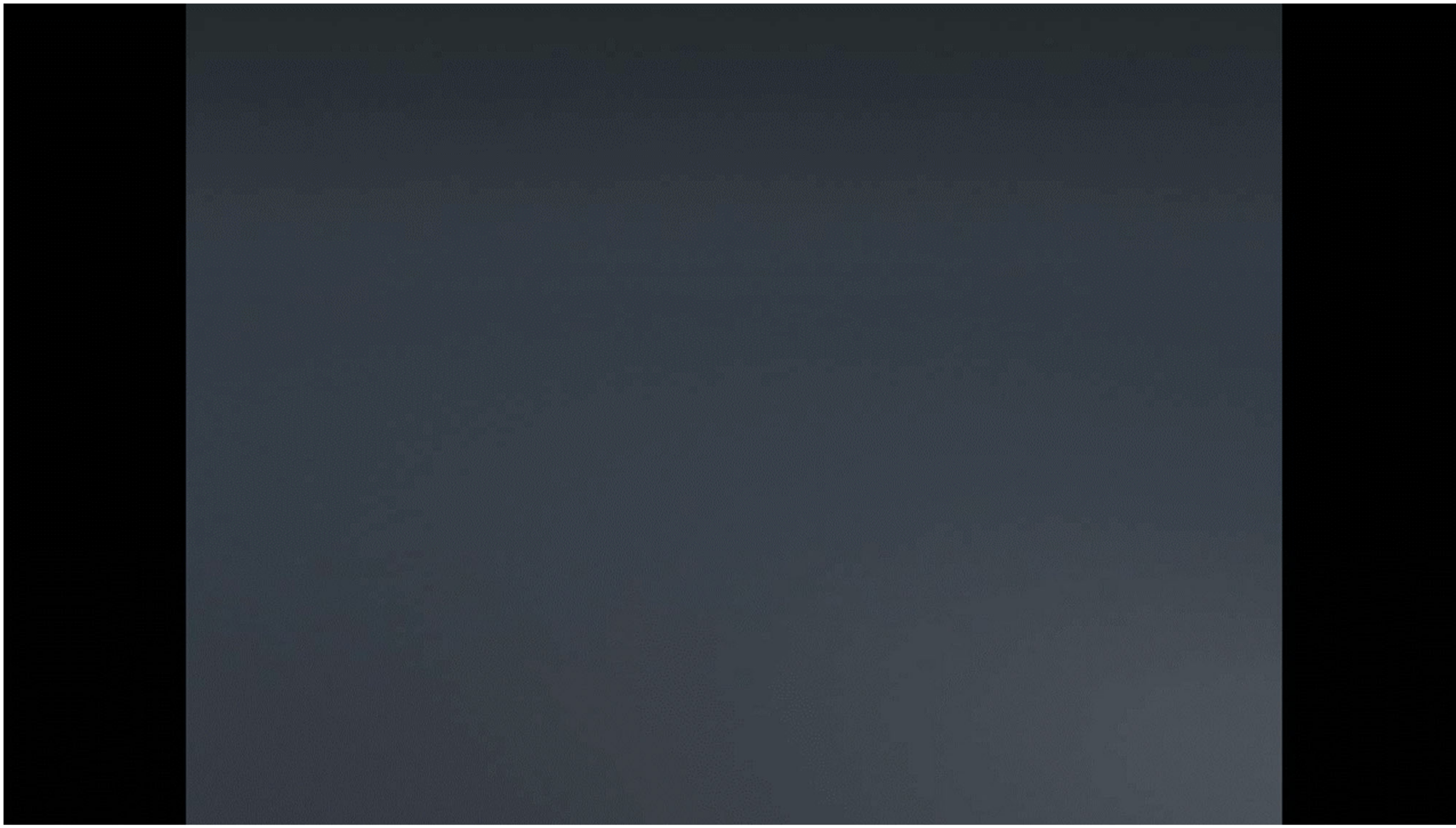
- This limitation can result in incomplete or distorted patterns, affecting the overall quality and performance of the final product.

- **Material Limitations**: Not all materials are compatible with nanoimprinting, which can limit its application in certain fields or require additional processing steps.

- **Throughput**: The process can be slower compared to other high-throughput fabrication methods like photolithography, especially for large-area patterns.

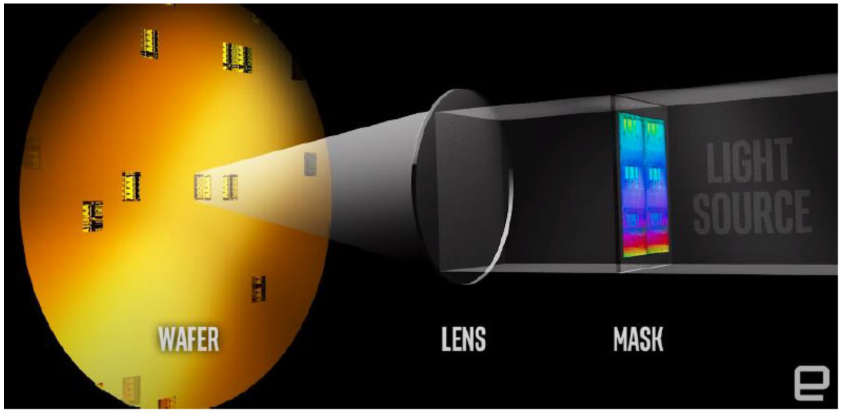
- **Temperature Sensitivity**: Some nanoimprinting techniques require high temperatures, which can limit the types of materials that can be used or affect the underlying substrates.

Nanoimprinting



EUV Photolithography

- **Extreme Ultraviolet Lithography (EUVL)** is the most advanced photolithography technique used in **semiconductor** manufacturing for patterning nodes below **7 nm and beyond**.



Source: Engadget, 2020

- Extreme Ultraviolet Lithography (EUVL) is a cutting-edge technology used in semiconductor manufacturing to produce smaller and more powerful microchips.

- Extreme ultraviolet lithography is an optical lithography technology using a range of extreme ultraviolet wavelength ~13.5 nm, to produce a pattern by exposing reflective photomask to UV light which gets reflected onto a substrate covered by photoresist.

- EUV light is strongly absorbed by air and most materials, which leads to unique design challenges.

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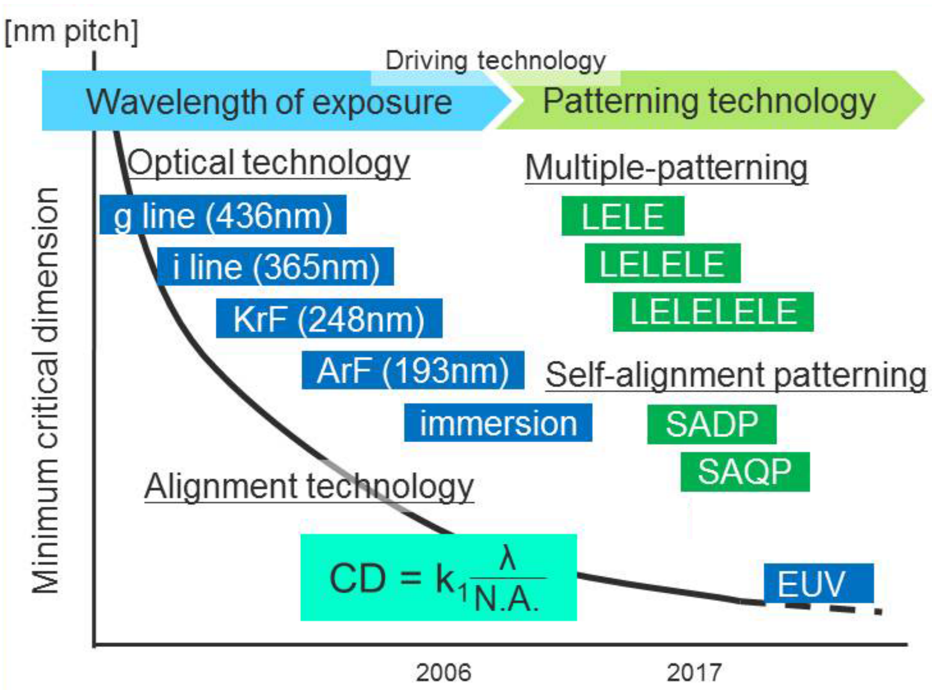
EUV Photolithography –wavelength Scaling

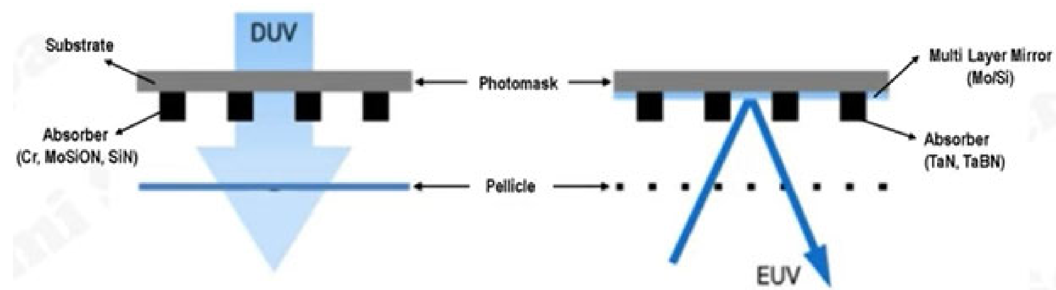
193nm Lithography at the Limit

The current state of the art has reached its physical limits. The structures on a chip are now many times smaller than the wavelength of light used to make them, and any more progress will require a big change.



Comparative Analysis: DUV and EUV Mask Technologies

Deep Ultraviolet (DUV, Deep UV) vs Extreme Ultraviolet (EUV) Reticles



ArF Lithography	EUV Lithography
Metal film on glass substrate, Step & scan, 4x Projection	
193nm	13.5nm
Atmospheric Pressure	Hydrogen Vacuum
Transmission & Absorption	Reflection & Absorption

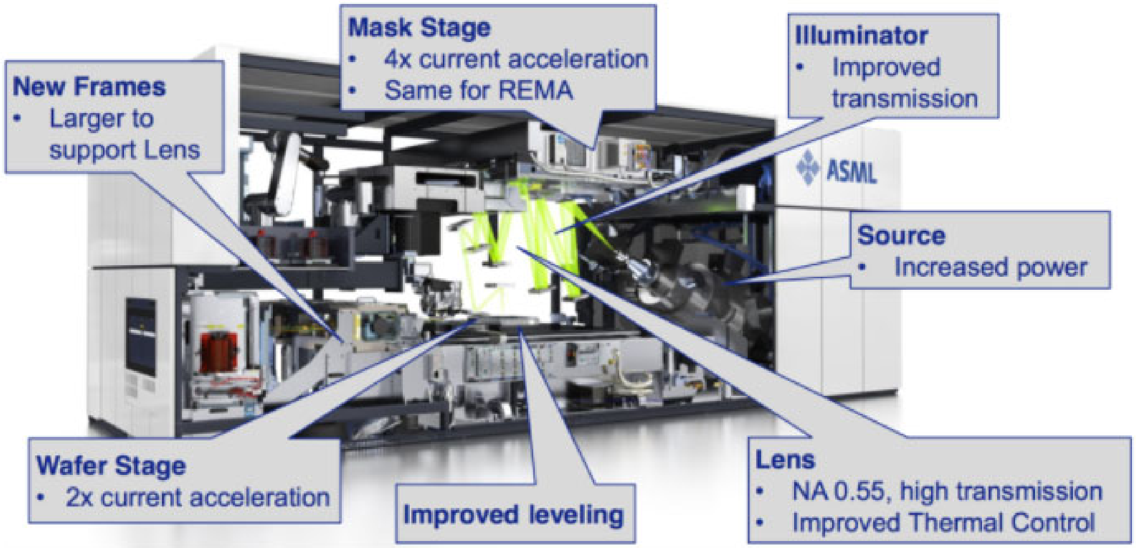
https://www.tel.com/rd/highlights/20171208_001.html

EUV Photolithography -System

And it's here: we see EUV - enabled chips in 2019

ASML

EUV up and running in High Volume Manufacturing

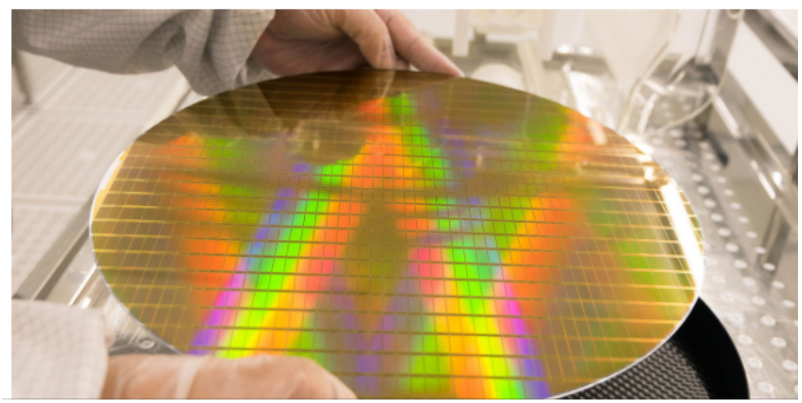


Courtesy: Semiconductor Engineering.

TSMC's True EUV Lithography Will Be On N5 Node For 2x Transistor Density

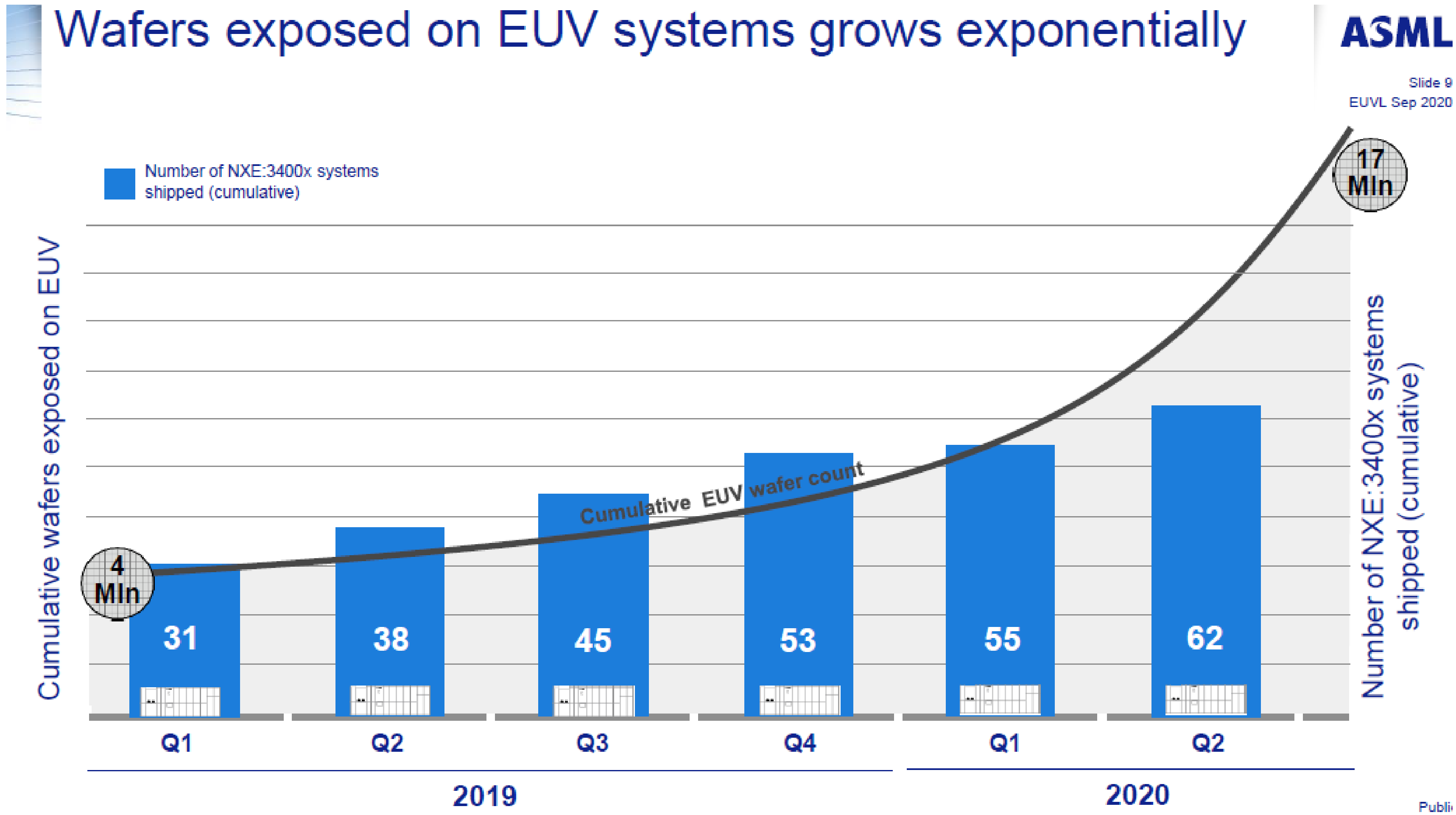
By Ramish Zafar

Aug 18, 2019 09:02 EDT



https://wccftech.com/tsmc-5nm-7nm-fabrication-details/

EUV Photolithography -System



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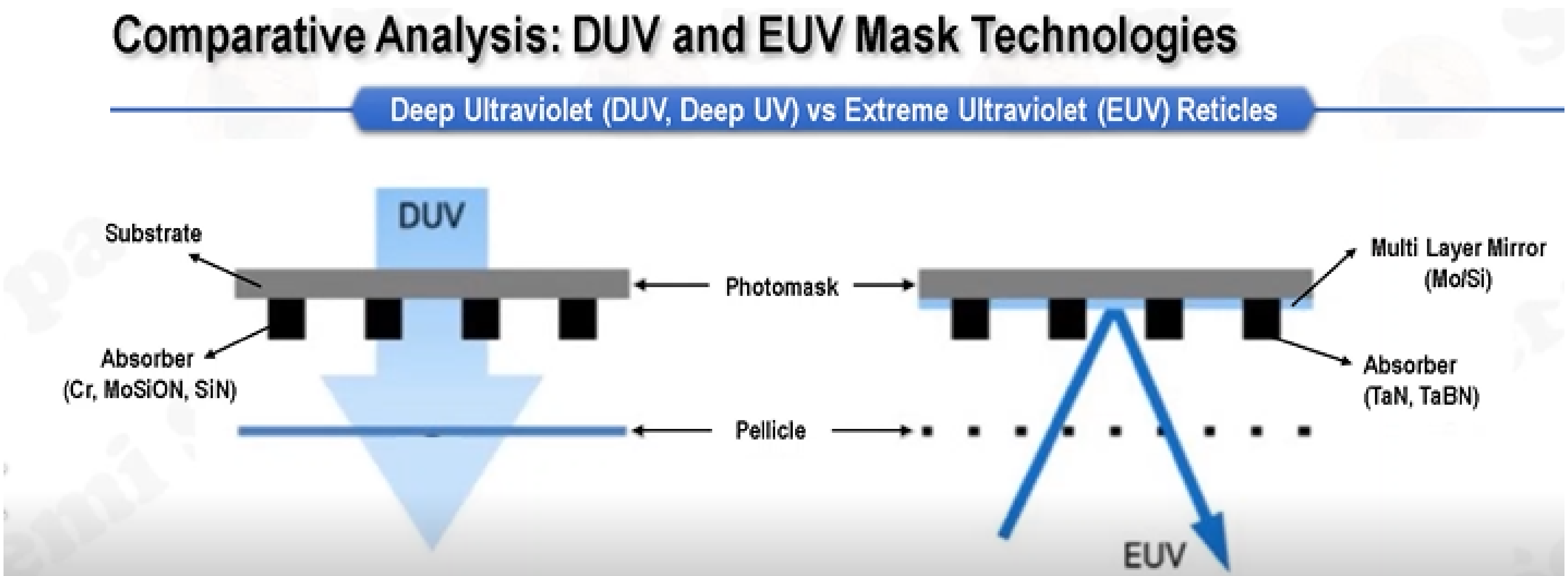
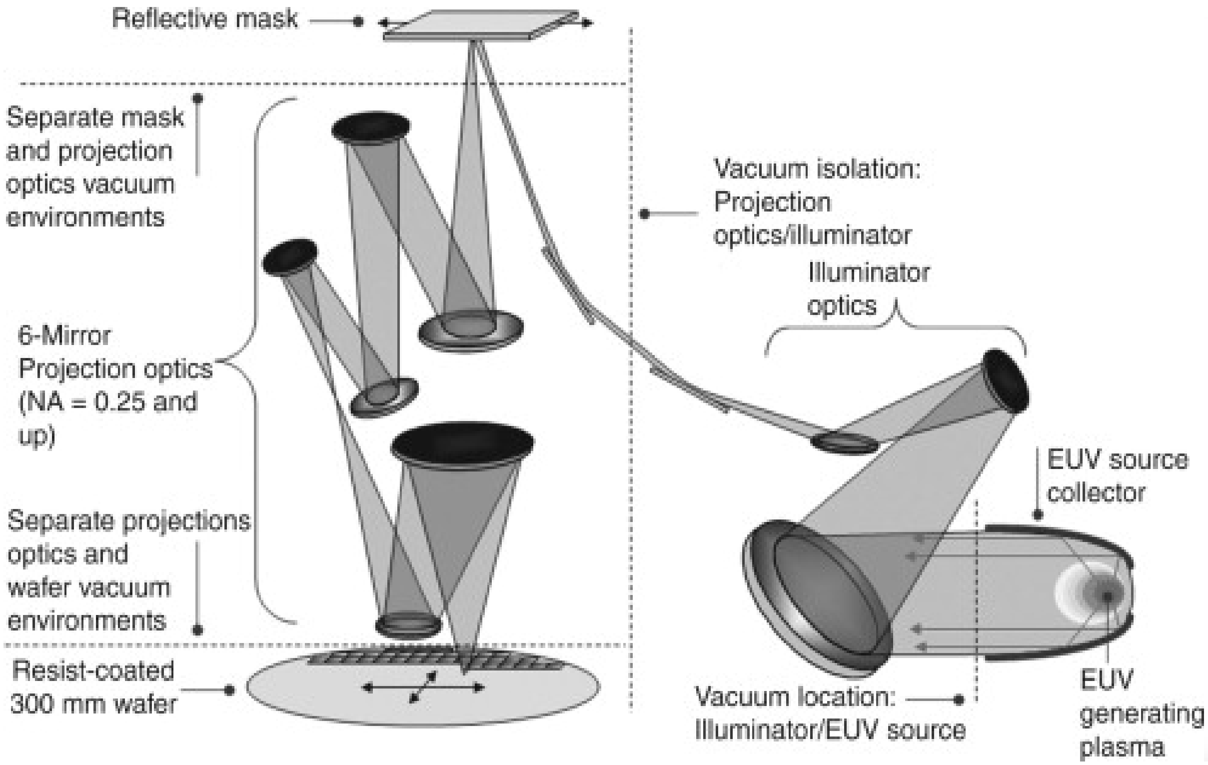
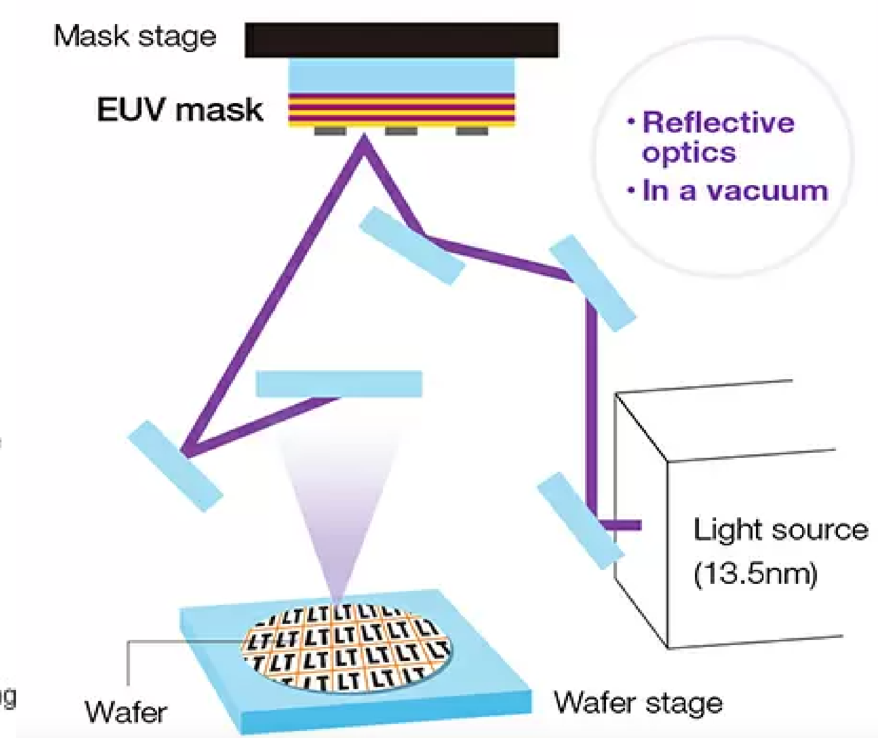


Ref: A Crappy Overview of EUVL-ASML

EUV Photolithography -System

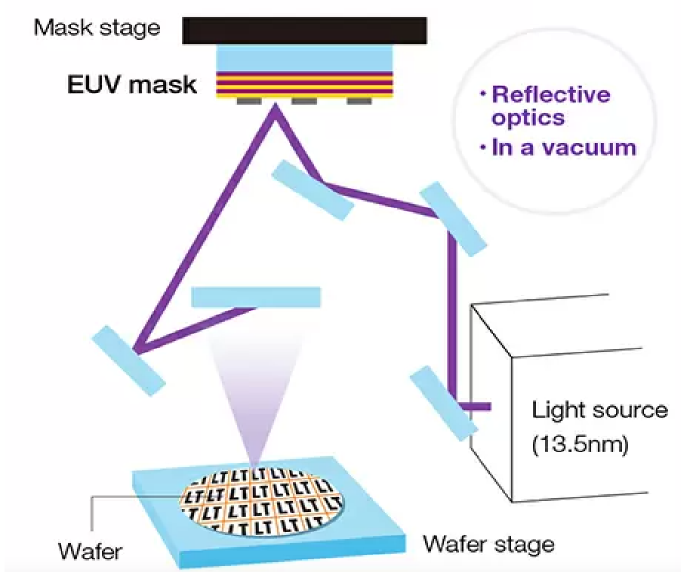




Ref: A Crappy Overview of EUVL-ASML

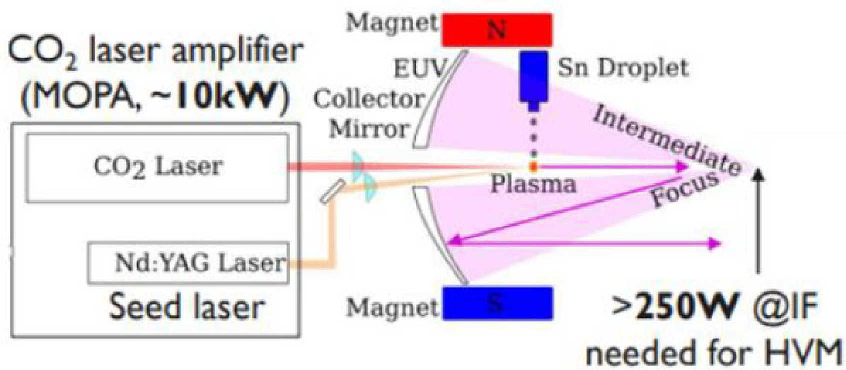
EUV Photolithography -System

Component	Function
EUV Light Source	Generates 13.5 nm light (plasma-based)
Collector Optics	Collects and focuses EUV light into the projection system
Illuminator	Uniformly illuminates the mask
EUV Mask (Reticle)	Reflective multilayer mask carrying the circuit pattern
Projection Optics	Series of mirrors that demagnify and project pattern onto wafer
Wafer Stage	Holds and moves wafer with nanometer accuracy
Vacuum Chamber	Entire system is in vacuum (EUV absorbed by air)

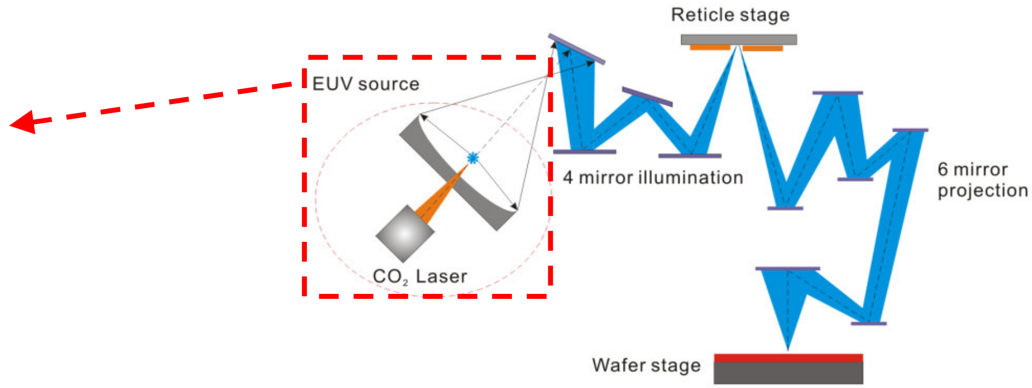


Ref: A Crappy Overview of EUVL-ASML

EUV Photolithography: Light source generation



Courtesy of Vu Luong, IMEC



Schematic diagram of an EUVL system.

A CO2 laser is a powerful, high-energy laser that uses a gas mixture with carbon dioxide to produce a beam of infrared light.

- Incident CO_2 and Nd:YAG laser beams hit a stream of microscopic droplets of molten tin (Sn) up to 50,000 times per second. This impact vaporizes the tin, creating a plasma that emits the desired 13.5 nm EUV light.

- EUV radiation takes place and focused by the collector mirror.

Ref: EUV Lithography: State-of-the-Art Review , N. Fu, et al., J. Microelectron. Manuf. 2, 19020202 (2019)

EUV Photolithography-Light collection and focussing

- Due to the strong absorption of 13.5nm EUV to any gases, the whole optical path must be fully reflective and enclosed in a vacuum environment, which is different from that of a deep UV scanner system.

- Moreover, in order to further reduce absorption and boost reflectivity, metal-based reflection mirrors is coated with alternating molybdenum and silicon (MoSi) layers with several nanometer in thickness, similar as the collector in EUV light source vessel.

Ref: EUV Lithography: State-of-the-Art Review , N. Fu, et al., J. Microelectron. Manuf. 2, 19020202 (2019)

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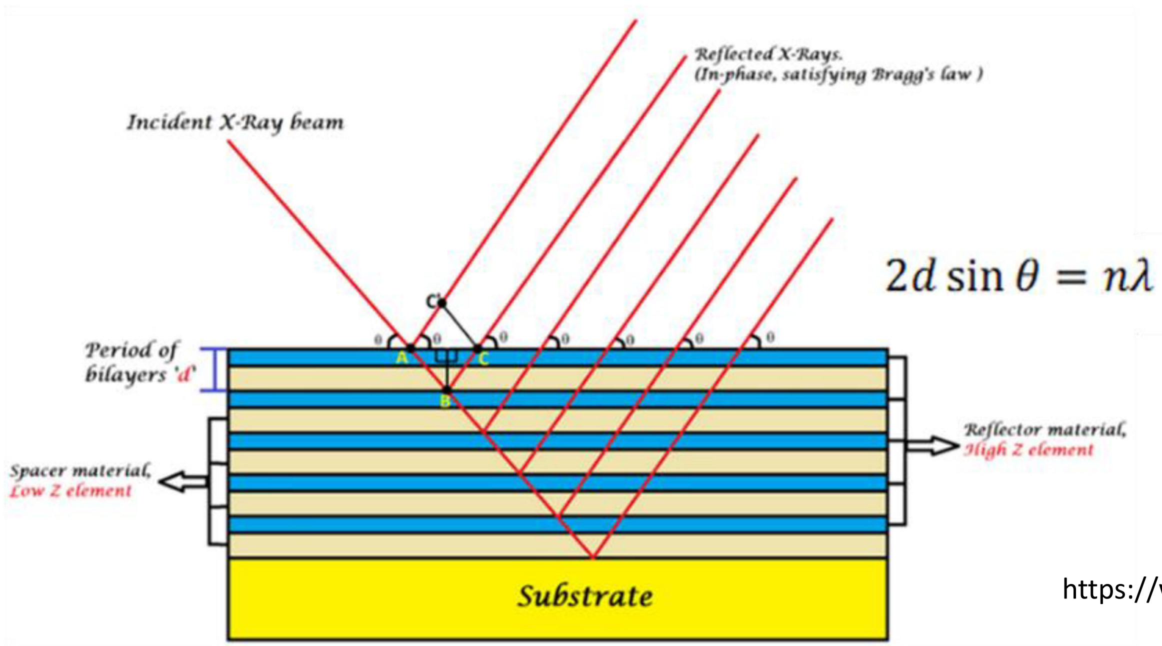


https://www.mdpi.com/2079-4991/11/11/2782

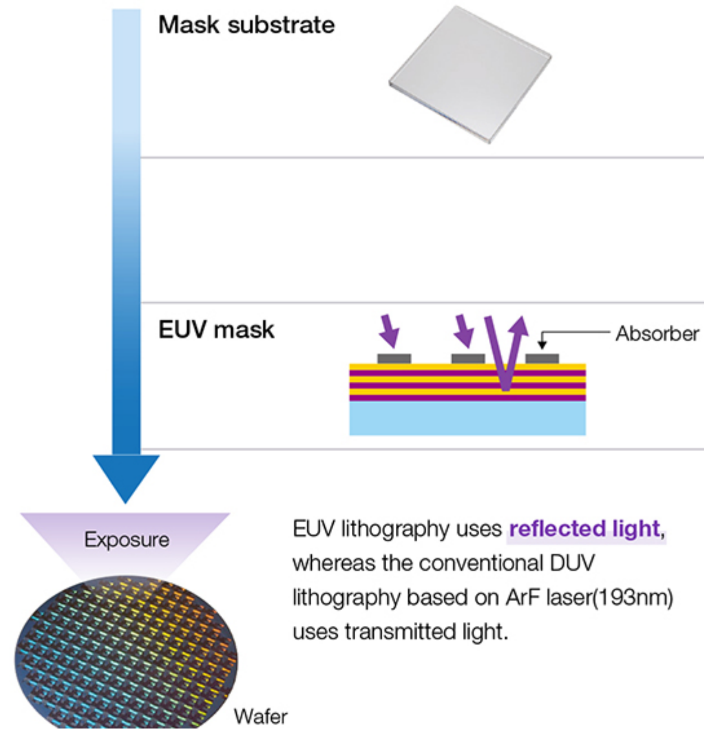
Ref: EUV Lithography: State-of-the-Art Review , N. Fu, et al., J. Microelectron. Manuf. 2, 19020202 (2019)

EUV Photolithography working-EUV Mask

- Similar to multiple coating on reflective mirrors in the vacuum system, 40 pairs of Si/Mo layers are deposited on the substrate of an EUV mask in order to get maximum reflection.

- A capping layer, a thin film of Ruthenium (Ru) to protect from oxidation and contamination, which would reduce reflectivity

- On top of the thermal emission layer, absorber layers will be patterned to form “clear” or “dark” features. Material selection of absorber is critical to maximize absorption and minimize reflection

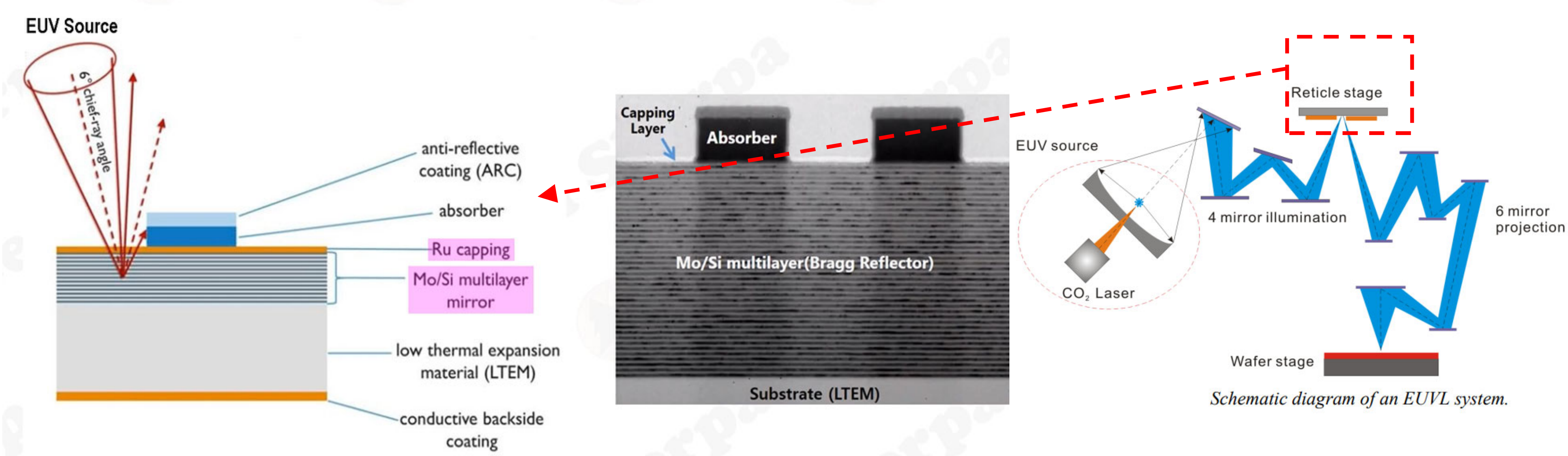


Ref: EUV Lithography: State-of-the-Art Review , N. Fu, et al., J. Microelectron. Manuf. 2, 19020202 (2019)

EUV Photolithography- Mask

- Between 40 and 50 alternating layers of silicon and molybdenum are deposited on the top of the substrate to act as a Bragg reflector that maximizes the reflection of the 13.5nm wavelength.

- This multi-layer reflector is then capped to prevent oxidation, typically with a thin layer of ruthenium.



LTEM (Low Thermal Expansion Material) substrate is a material, often a type of glass, that expands very little when heated

EUV Photolithography- Mask

- The EUV Mask is a patterned multilayer mirror
- A Bragg reflector is a multilayer structure of alternating high and low refractive index materials that reflects a specific band of wavelengths
- Bragg reflector, which uses multilayer interference to achieve high reflectivity (up to about 70%) at the specific 13.5 nanometer EUV wavelength
- The mask must be perfect to ensure defects are not transferred to the wafer, affecting yields
- Multiple inspection steps required throughout the manufacture and lifetime of the mask

Ref: Extreme Ultraviolet Light Sources Supporting Next-Generation Lithography, Sam Gunnel, ENERGETIQ, 2020

EUV Photolithography- Photo resist

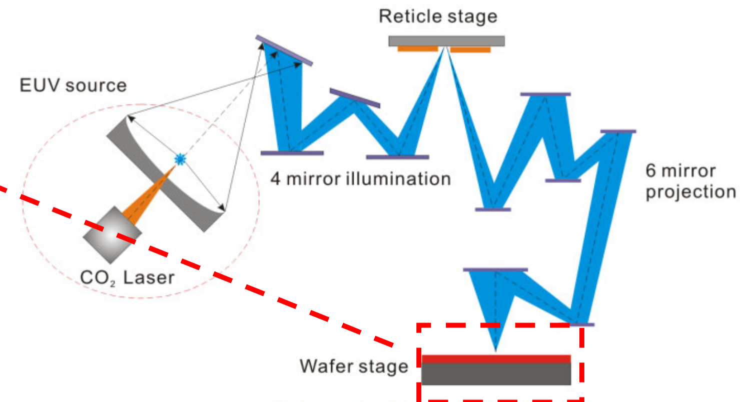
EUV Photoresist

- The **photoresist** is a thin polymer film that captures the EUV pattern projected from the mask.
- It must convert EUV photons (92 eV) into **chemical reactions** that define the pattern upon **development**.

Photon Absorption

- EUV photons- high energy (92 eV) and can generate multiple **secondary electrons (~20–80 eV)** in the resist causing **chemical changes** within the resist film.

EUV Photolithography- Photoresist



Schematic diagram of an EUVL system.

Depending on the resist type:

- **Positive Resist:** Exposed regions become more soluble.

Development

•The wafer is immersed in a **developer solution** (usually TMAH — tetramethylammonium hydroxide).

•The soluble parts are removed, revealing the pattern.

EUV Photolithography –Other Challenges

- Driven by >20kW CO_2 laser
- Prohibitively expensive to own and operate
- 250W EUV at intermediate focus
- Not suitable for research or metrology applications

Ref: Extreme Ultraviolet Light Sources Supporting Next-Generation Lithography, Sam Gunnel, ENERGETIQ, 2020

EUV Photolithography –Challenges

Technical and engineering challenges

- **Light source:** EUV systems require highly complex and powerful light sources, which can have reliability issues and are expensive to maintain.
- **Optics:** The reflective mirrors must be incredibly precise, with layers just a few molecules thick and requiring accuracy at the angström level. Contamination of these mirrors is a major issue that necessitates a vacuum environment and specialized hydrogen use to remove contaminants.
- **Mask defects:** EUV masks are prone to defects, and defects are much harder to detect and repair than in previous technologies, which can degrade chip quality.

EUV Photolithography –Challenges

Material and process challenges

- **Photoresists:** Developing photoresists that are sensitive enough for high throughput while also being strong enough to prevent pattern collapse during rinsing is difficult. Some resists suffer from stochastic failures where random photon placement creates defects.
- **Environmental control:** The entire beam path must be in a vacuum. The use of hydrogen to manage contaminants adds a layer of complexity and safety concerns, even though it can be managed to be non-flammable.
- **Photon shot noise:** For smaller features, there are fewer photons, which can lead to random variations in the exposure dose, a phenomenon known as shot noise that can cause pattern defects.

EUV Photolithography –Challenges

Manufacturing and business challenges

- **Equipment cost:** EUV machines are extremely expensive, making adoption difficult for smaller manufacturers.
- **High-volume manufacturing (HVM):** Scaling laboratory techniques to the rigorous demands of high-volume production is an ongoing challenge.
- **Infrastructure and business models:** The industry faces significant challenges in building the supporting infrastructure and developing new business models to share costs between equipment suppliers and device manufacturers.

EUV Photolithography –Challenges

Summary of Key Bottlenecks

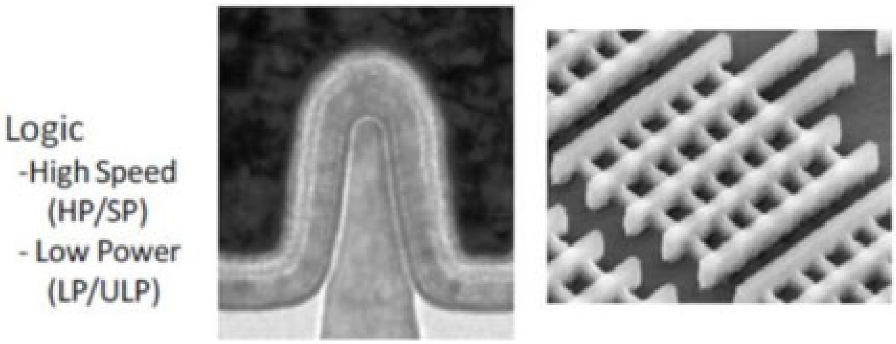
Category	Challenge	Impact
Source	Low conversion efficiency	Low throughput
Optics	Mirror contamination	Reduced reflectivity
Mask	Defects, pellicle heating	Print errors
Resist	Stochastic noise, LER	Critical dimension variation
Etch/Process	Resist thinning, collapse	Pattern fidelity
Overlay	Multi-layer alignment	Yield loss
Cost	Equipment, maintenance	Limited adoption

EUV Photolithography –Expectations

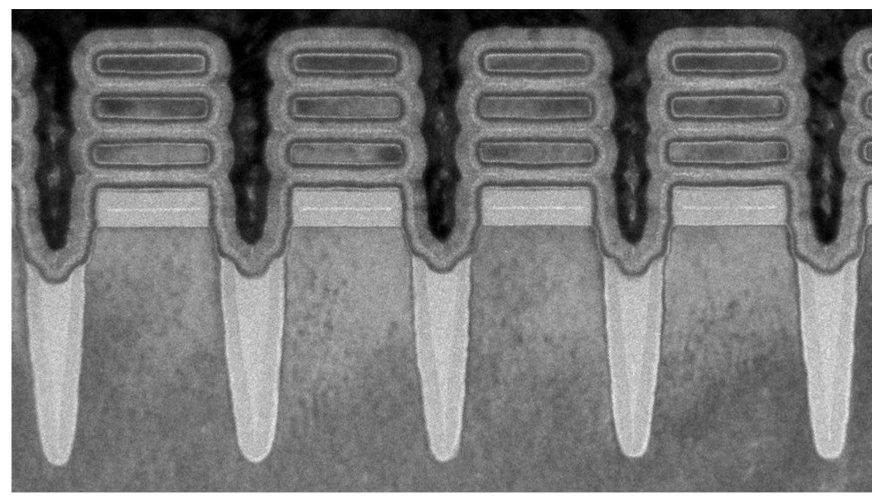
- Lower patterning cost
- More efficient use of EUV
- Faster time to market
- Ramping to higher yield

EUV Photolithography

EUV Photolithography -some patterns of FinFET



https://wccftech.com/tsmc-5nm-7nm-fabrication-details/



IBM: https://analyticsindiamag.com/a-small-chip-for-ibm-a-giant-leap-for-semiconductor-industry/

EUV Photolithography –Summary

- EUV Lithography has entered high volume manufacturing for certain layers, but 193nm multi-patterning is still the primary production method
- Researchers and metrologists continue to need cost-effective sources of EUV photons to address EUV roadmap and infrastructure requirements
- Each type of EUV source has its own advantages and disadvantages: there is no one-size-fits all solution
- Few systems are already installed with consistent operation over the last few years

Ref: Extreme Ultraviolet Light Sources Supporting Next-Generation Lithography, Sam Gunnel, ENERGETIQ, 2020

Simulation study of MOS capacitor and MOSFETs using Nanohub

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Simulation study of MOS capacitor

- C-V characteristics
- Correlation to experimental results

MOS cap tool: https://nanohub.org/resources/moscap

Intro to MOS-Capacitor Tool:

https://nanohub.org/resources/mosctool

Simulation study of MOS capacitor and MOSFETs using Nanohub

Simulation study of MOSFETS

- Id-Vg and Id-Vd characteristics
- Effect of dielectric, doping and other parameters on MOSFET

working

MOSFET: https://nanohub.org/tools/mosfet/session?

sess=2513871

- https://nanohub.org/tools/mosfetsat/session?sess=2513873

Course Project 1: Discussion